

Physics A (H156, H556)

Exploring - Diagnostic methods in medicine ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is 53.
- The total number of marks may take into account some 'either/or' question choices.

- i. Explain what is meant by a *tracer*.

----- [1]

- ii. $^{99}_{43}\text{Tc}^m$ only emits gamma radiation.

Give **two** advantages of using a tracer which only emits gamma radiation.

1 -----

2 -----

----- [2]

- 2(a) Radiographers commonly use molecules containing fluorine F-18 as tracers in positron emission tomography (PET) scanning.

Fluorine has a proton number of 9.

F-18 decays to oxygen (O) by β^+ decay.

Write the equation for the decay of a nucleus of F-18 using nuclear notation.

[2]

- (b) The β^+ particle (positron) produced travels only a short distance in the patient before it meets an electron and is annihilated.

Calculate the wavelength λ of gamma photons produced.

$\lambda = \dots\dots\dots\text{m}$ [3]

- (c) X-rays and gamma-rays are produced by different physical processes.

Briefly describe both processes.

----- [2]

- (d) F-18 has a half-life of 109.7 minutes.

Explain the advantage that this has for the patient but the disadvantage that this has for the radiographers.

----- [3]

3 Technetium-99m (Tc-99m) is a metastable isotope used in medical diagnosis.

Which ionising radiation does Tc-99m emit?

- A alpha
- B beta-minus
- C beta-plus
- D gamma

Your answer

[1]

4(a) A gamma camera has several important components including a collimator, scintillator and photomultiplier tubes.

Suggest why the collimator needs to be long and narrow.

----- [1]

(b) State the function of the scintillator.

----- [1]

(c) In a single photomultiplier tube, a photon of light produces a $0.32 \mu\text{A}$ pulse of current for a duration of 1.2 ns .

Calculate the number of electrons responsible for this pulse of current.

number of electrons = [2]

(d) State one diagnostic application of a gamma camera.

----- [1]

- 5 The medical tracer fluorine-18 is used in positron emission tomography (PET).
Fluorine-18 is a beta-plus emitter with a short half-life.

Describe how the fluorine-18 nuclei are located in a patient using a PET scanner.

[4]

- 6(a) The medical tracer technetium-99m is used in imaging organs such as the brain.

A gamma-camera uses powerful computers and sophisticated software to produce three- dimensional images of the patient's organ.

Name and describe the remaining three main components of the gamma camera.

[3]

(b) Explain the advantages of using technetium-99m for this purpose.

----- [2]

7 The Doppler effect is used to measure the speed of blood flow in arteries.

Which medical diagnostic method uses this technique?

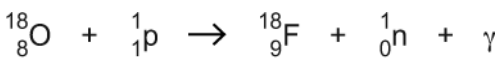
- A barium scan
- B CAT scan
- C PET scan
- D ultrasound scan

Your answer

[1]

8

The nuclear reaction below shows how the isotope of fluorine-18 ($^{18}_9\text{F}$) is made from the isotope of oxygen-18 ($^{18}_8\text{O}$).



The oxygen-18 nucleus is **stationary** and the proton has kinetic energy of $0.25 \times 10^{-11} \text{ J}$.

The binding energy of the $^{18}_8\text{O}$ nucleus is $2.24 \times 10^{-11} \text{ J}$ and the binding energy of the $^{18}_9\text{F}$ nucleus is $2.20 \times 10^{-11} \text{ J}$. The proton and the neutron have zero binding energy.

i. Explain why a high-speed proton is necessary to trigger the nuclear reaction shown above.

----- [2]

ii. Estimate the minimum wavelength λ of the gamma ray photon (γ).

$\lambda =$ ----- m [3]

iii. Fluorine-18 is a positron emitter.

Name a medical imaging technique that uses fluorine-18 and state one benefit of the technique.

----- [2]

9 A contrast material is used while taking an X-ray image of a patient.
Which statement is correct?

- A Iodine is a contrast material.
- B Technetium is a contrast material.
- C A contrast material must have a short half-life.
- D A contrast material is used for acoustic matching.

Your answer

☐

[1]

Describe how a PET scanner is used to locate an area of increased activity within the patient.

[4]

Calculate the time t in minutes for the activity of the radiopharmaceutical to decrease to 30% of its initial activity.

$t =$ _____ minutes [3]

(c) PET scanners are not available in all hospitals. This is because fluorine-18 requires expensive on-site particle accelerators and fluorine-18 has a very small 'shelf-life'.

Suggest the impact this may have on the treatment and diagnosis of patients in the country.

-----[1]

- 11 A gamma camera is connected to a computer and a display. Sophisticated software is used to produce a quality scan (image) of the patient.

i. Briefly describe the function of the collimator, scintillator and photomultiplier tubes in a gamma camera.

[3]

- ii. Fig. 23 shows two types of lead collimator tubes T and S available for a gamma camera.



Fig. 23

Tube **T** is thin and long. Tube **S** is broad and short.

Discuss which type of tube would be more suitable in a gamma camera.

[2]

12 Which is **not** a component used in a gamma camera?

- A X-ray tube
- B collimator
- C computer
- D photomultiplier tubes

Your answer

[1]

13 PET scanners are expensive because they require a near-by or on-site particle accelerator that produces fluorine-18. Discuss the ethical issues this raises in the treatment of patients.

[2]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1		i	Material (introduced into the body and) used so its position in the body can be detected / function of organ can be determined	B1	Allow “used to produce images” for “detected” Do not allow “trace” for “position detected” Ignore treatment <u>Examiner’s Comments</u> There are many possible answers to this question, but it was felt that the key point being looked for was the idea that a location in the body could be identified. General uses of tracers was not specific enough and several candidates simply stated that the tracer could be traced. Only around a quarter of candidates gave a suitable response to this question.
		ii	(Very) little ionisation (Most can) pass through the body tissue / can be detected outside / highly penetrating	B1 B1	Allow less ionising / least ionising Do not allow “zero ionisation” Allow other radiation would be absorbed within the body <u>Examiner’s Comments</u> This relatively simple question was correctly answered by the majority of candidates. There was no additional detail required apart from the two simple facts although it was good to see many candidates putting these into context. Candidates should be reminded that in a question like this, only two responses should be given. Including a third runs the risk of a contradiction. Common misconceptions included the idea that the half-life of a source of gamma radiation was shorter (or longer) than alpha and beta, and that gamma radiation was ‘safer’.
			Total	3	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
2	a	${}^{18}_{9}\text{F} \rightarrow {}^{18}_{8}\text{O} + {}^0_1\beta^+ (+\nu)$	C1 A1	Correct fluorine isotope Correct equation ALLOW numbers written to right of symbol ALLOW $e / e^+ / \bar{e} / \beta$ for positron symbol with correct numbers IGNORE gamma in products ALLOW ${}^0_0\nu$ for neutrino ALLOW ν_e for neutrino <u>Examiner's Comments</u> Most candidates will have gained the first mark for the correct isotope of fluorine. Those who didn't likely reversed the positions of the nucleon and proton number. It was noticeable that a large number of candidates were unable to balance the equation and while they had the correct values for the positron, the oxygen isotope was incorrect.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	$E = mc^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2$ $= 8.2 \times 10^{-14} \text{ (J)}$ $E = hc / \lambda = 8.2 \times 10^{-14} = 6.63 \times 10^{-34} \times 3 \times 10^8 / \lambda$ $\lambda = 3.00 \times 10^8 \div 1.2367 \times 10^{20} = 2.4 \times 10^{-12} \text{ (m)}$	C1 C1 A1	Working leading to evaluation of E ALLOW factor of 2 leading to 1.6×10^{-13} (J) NOT use of $\lambda = h/mv$ where v is 3×10^8 alone XP ALLOW use of $\lambda = hc/mc^2$ or $\lambda = h/mc$ ALLOW Use of electron mass = 0.511 MeV giving 8.2×10^{-14} (J) Correct to at least 2 significant figures Answer of 4.9×10^{-12} or 1.2×10^{-12} allow 2/3 due to incorrect accounting for factor of 2 <u>Examiner's Comments</u> Well over half of the candidates were able to score 2 or more marks on this question. A great deal scored 2 marks due to the inclusion of a factor of two, which was then not removed (or vice versa). Some candidates used the formula for kinetic energy – which by coincidence may give the correct answer – however this was not credited, nor was the more common use of the de Broglie formula. Examiners sometimes had a difficult decision on this question, whether the response was a physics error or not, and if in doubt a candidate would be awarded the marks.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	c	X-rays formed when <u>electrons</u> (in an atom) de-excite Gamma rays come from the decay of <u>nuclei</u> (in unstable isotopes)	B1 B1	ALLOW X-rays may be produced by acceleration / deceleration of (fast moving) electrons / X-rays are produced when (fast moving) electrons are incident on a metal target / X-rays may be produced when electrons lose kinetic energy ALLOW gamma rays come from (the decay of) radioactive <u>nuclei</u> / gamma rays come from the <u>nucleus</u> of unstable atoms / gamma rays come from the de-excitation of <u>nuclei</u> / gamma rays come from <u>annihilation</u> of particle-antiparticle (pairs) / IGNORE gamma from fission <u>Examiner's Comments</u> The main principle behind this question was to distinguish between the X-ray production by electrons and gamma ray production by nuclei. Fairly specific details were required for each, although a wide range of answers were accepted. This question discriminated well among the candidates with roughly equal fractions getting 0, 1 and 2 marks.
	d	The half-life is short Advantage: Exposure of the patient to <u>ionising</u> radiation is kept as low as possible. Disadvantage: (Radiographers have a) short time to scan/diagnose the patient	B1 B1 B1	ALLOW activity is high IGNORE it decays quickly ALLOW F18 has to be manufactured on site before use / high activity means exposure is high during handling IGNORE short time to treat the patient <u>Examiner's Comments</u> Although apparently a simple set of ideas, the detail required for this question meant that only a small fraction of candidates were able to gain full marks. It was necessary to state that the half-life was (relatively) short, which many of the stronger candidates did not do. The advantage to the patient requires the use of the term 'ionising' as this is the fundamental issue with the radiation. Similarly, candidates had to make it clear that the disadvantage was due to the short time to carry out the scan rather than just a statement of needing to work quickly.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	10	
3			D	1	<u>Examiner's Comments</u> The majority of candidates were able to recall the correct emission, with the remaining responses being fairly well split across the other options, suggesting some guess work was involved.
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
4	a	Allows only those gamma rays / waves / photons travelling along axis (of the collimator) to get through (and reach scintillator)	B1	Allow less fuzzy / clear image Allow all the gamma rays / waves / photons are parallel / in the same direction (to each other) Allow <u>absorbs</u> those gamma rays / waves / photons not parallel (to the axis of collimator) Allow so the photons are travelling perpendicular to the scintillator Do not allow the gamma rays are travelling vertically, unless it is clear the collimator is also vertical <u>Examiner's Comments</u> This question was well answered by around half of the candidates who appreciated that the long, narrow collimator would allow only gamma rays which were parallel to each other to be received by the scintillator. This can be expressed in a number of ways and marks were given to candidates who were able to state this using alternative descriptions. A common confusion appeared between using the terms perpendicular and parallel, with some candidates incorrectly stating that the collimator allowed photons perpendicular to the tubes to pass. Essentially, the collimator allows for a clearer image to form and so this is an acceptable response.
	b	Turn gamma (photons) into (many photons of) light	B1	Ignore reference to rays / waves Ignore reference to flash <u>Examiner's Comments</u> Around half of the candidates were able to correctly explain that the gamma photons produced visible light in the scintillator. Several candidates thought that the scintillator produced electrons or a voltage when the gamma photons were incident on it instead. Many candidates gave the extra correct information that many visible photons are produced from one gamma photon, although this was not a required detail on this occasion.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		number of electrons = $\frac{0.32 \times 10^{-6} \times 1.2 \times 10^{-9}}{e}$ number of electrons = 2400	C1 A1	Ignore any POT error for C1 mark <u>Examiner's Comments</u> This calculation was done well by a large majority of candidates. There were few errors in the unit prefixes, with the most common one being 1.2 ns as 1.2×10^{-12} s. A small number of candidates calculated the charge correctly, but then took it no further to determine the number of electrons.
	d		Any sensible <u>diagnostic</u> suggestion, e.g. <u>detection</u> of cancer / scans of (named) organ / scans of tissue / bone scans / observing functionality of (named) organ	B1	Not medical <u>treatment</u> e.g. radiotherapy Not body scan Ignore PET scanner Do not allow CAT scan <u>Examiner's Comments</u> The key word here was “diagnostic” and responses were expected to state any reasonable diagnostic use of a gamma camera and a wide variety of responses were given and accepted. Common responses included checking for brain tumours and observing kidney failure. Although a gamma camera may be used in a PET scanner, this alone was not acceptable as it does not explain the diagnostic use. Candidate should be encouraged to write in a clear sentence structure as simple responses such as “cancer” cannot be given marks.
		Total		5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5			The positrons / beta-plus particles <u>annihilate</u> electrons (within the patient)	B1	Allow ‘two gamma rays’ instead of ‘two gamma-photons’
			Two gamma-photons are produced	B1	Allow gamma symbol
			these (photons / rays) travel in opposite directions	B1	Allow delay time
			The difference in the arrival times at the detectors is used to locate the point of annihilation / nuclei	B1	<u>Examiner’s Comments</u>
					This question covers part of the diagnostic methods in the diagnostic methods in medicine topic from module 6. This question is directly from the learning outcome 6.5.2(e). Candidates were expected to describe the basic principles of the PET scanner and use this to describe how to locate fluorine-18 nuclei. This question was poorly answered by many candidates and over half scored zero marks. Of that, there were a significant number who gave no response. However, there were some excellent responses that demonstrated a genuine understanding of the process. Several candidates thought that the beta-plus particles were electrons and from that point would have difficulty in making progress. The short half-life seemed to cause some confusion, and several candidates felt that this was a key part of the question. Of the candidates who explained the principles of operation clearly, some were unable to explain that it is a time difference between the arrival of the gamma rays at the detectors that is used for location.
Total			4		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a		Collimator: Allows gamma (photons) parallel to the axis of the tubes to pass through Scintillator: gamma (photons) produces (many) light (photons) Photomultiplier (tubes): light (photons) produces electrons / current / electrical pulse / p.d. / signal	B1 B1 B1	Ignore any other components named / described Allow photon / waves / rays Allow idea of tubes allowing the gamma (photons) to travel in the same direction Allow crystal (or named crystal) for scintillator Allow high-energy photons produce (many) low-energy photons
	b		Emits gamma (photons / radiation / waves / rays) Any one from: (Diagnosing the) function of organ Detecting tumour Small half-life (Gamma rays) can be detected outside body / passes through patient / least ionising Position of tracer located	B1 B1	Not injected into a patient / non-invasive Allow for half-life is a few hours
			Total	5	
7			D	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8		i	Proton is repelled (by nucleus) (High-speed) proton can get close to (oxygen) nucleus	B1 B1	Allow 'proton can experience the strong (nuclear) force' Not 'collide / hit nucleus'
		ii	$E = [0.25 - (2.24 - 2.20)] \times 10^{-11} \text{ (J) or } 0.21 \times 10^{-11} \text{ (J)}$ $\lambda = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{0.21 \times 10^{-11}} \text{ (Any subject)}$ $\lambda = 9.5 \times 10^{-14} \text{ (m)}$	C1 C1 A1	Allow 2 marks for 6.9×10^{-14} ; $E = 0.29 \times 10^{-11}$ used Allow 1 mark for a value correctly calculated based on any other incorrect value for E (e.g. $8(.0) \times 10^{-14}$ for $E = 0.25 \times 10^{-11}$ and $5(.0) \times 10^{-13}$ for $E = 0.04 \times 10^{-11}$)
		iii	Used in PET (scans) Any one from: Used to diagnose function of organ / brain / body Detection of cancer / tumour Non-invasive / no surgery / no infection 3D (image)	M1 A1	Enter text here.
			Total	7	
9			A	1	Examiner's Comments The majority of the candidates did get the correct answer A. A significant number of candidates opted for C, confusing contrast material with medical tracers.
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
10	a	The patient is surrounded by (gamma) detectors or Increased activity is where F-18 accumulates (AW)	B1	Allow 'diametrically opposite detectors'
		The positrons (from the F-18) <u>annihilate</u> electrons (inside the patient)	B1	
		Each annihilation produces two gamma photons travelling in <u>opposite</u> directions	B1	Not gamma rays / radiation
		The arrival times are used to locate position (of increased activity)	B1	Allow 'delay time'
				Examiner's Comment Most candidates scored two or more marks for their description of the PET scanner. Most candidates knew that the annihilation of positrons and electrons was central to the scanning technique. A small number of candidates either confused the PET scanning with CAT scanning or assumed that the gamma detectors were monitoring the emission of positrons from the patient.
	b	$\lambda = \ln 2 / 110$ or $6.3 \times 10^{-3} \text{ (min}^{-1}\text{)}$	C1	Allow $1.05 \times 10^{-4} \text{ (s}^{-1}\text{)}$
		$0.30 = e^{-6.3 \times 10^{-3} t}$		This is the same as $0.30 = e^{-1.05 \times 10^{-4} t}$
		$t = \frac{\ln(0.30)}{-6.3 \times 10^{-3}}$	C1	Note: This mark is for a ln expression (any subject)
		$t = 190 \text{ (minutes)}$	A1	Allow 2 marks for $1.15 \times 10^4 \text{ (s)}$ as the final answer
				Examiner's Comment This was not an easy question. It required knowledge and understanding of activity, decay constant and natural logs. It is good to report that most of the candidates produced immaculate answers. The common mistakes made were: • Using either $\ln(1/3)$ or $\ln(0.70)$ rather than $\ln(0.30)$ in the calculations. • Assuming the decay was linear rather than exponential.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		Any sensible suggestion, e.g. 'post-code' lottery, some patients may not get the treatment because of where they live, longer waiting lists, etc.	B1	Examiner's Comment Almost all candidates gave a plausible suggestion in this last question in the paper. It is good to report that physicists are mindful of the impact of science on society.
			Total	8	
11		i	Collimator: Only gamma rays / photons travel along the axes of lead tubes are detected. (AW)	B1	
		i	Scintillator: A gamma ray photon produces thousands / many photons of (visible) light. (AW)	B1	
		i	Photomultiplier: An electrical pulse is produced from each photon of visible light entering a photomultiplier tube. (AW)	B1	
		ii	The long and thin tubes would be suitable because gamma photons over smaller spread of angles / area of patient would be detected. (AW)	B1	
		ii	This would produce a clearer / sharper / less blurred image (scan) of the patient. (AW)	B1	
			Total	5	
12			A	1	
			Total	1	
13			Doctors have to make difficult decisions about who can and cannot have a PET scan.	B1	
			Some patients will miss out on PET scans because of their location / not all patients will have access to the scans.	B1	
			Total	2	